

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Ashwin V-(192424345) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V Ashwin

Annexure A

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than **1,200 employees** use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's **Wireless Local Area Network (WLAN)**. The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use **4G/5G Wireless Wide Area Network (WWAN)** services to access the company's servers, customer databases, and business applications from remote locations.

During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an **Asynchronous Transfer Mode (ATM) network** to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size **53-byte cells** and establishes **Virtual Circuits (VCs)** to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.
3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a **Multiprotocol Label Switching (MPLS) network** to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences.

Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.
5. A leading online education platform conducts live classes and online examinations for more than **15,000 students** across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed **10 Gbps** connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

CSA0713 CO2 AT3 – Debugging & Optimization Task Answers

Question 1: WLAN and WWAN Network Issues

Root Causes:

- Overloaded wireless access points.
- Poor access point placement causing weak signals.
- Improper roaming configuration between floors.
- Bandwidth congestion during peak hours.
- Weak 4G/5G coverage in rural areas.
- Use of outdated security protocols.

Debugging Process:

- Analyze WLAN traffic using monitoring tools.
- Check access point utilization.
- Perform wireless site surveys.
- Test roaming performance.
- Measure WWAN signal strength.
- Audit security settings.

Optimization Techniques:

- Deploy additional access points.
- Enable load balancing.
- Configure seamless roaming.
- Upgrade to Wi-Fi 6/Wi-Fi 6E.
- Use signal boosters for WWAN.
- Implement WPA3 security.

Expected Outcome:

Improved coverage, reduced disconnections, faster speeds, better video conferencing, and stronger security.

Question 2: ATM Network Performance Issues

Root Causes:

- Congested Virtual Circuits (VCs).
- High ATM switch utilization.
- Improper bandwidth allocation.
- Incorrect ATM service categories.
- Poor QoS management.

Debugging Process:

- Monitor ATM switches.
- Analyze VC utilization.
- Check cell loss and delay.

- Verify QoS configurations.
- Identify overloaded traffic classes.

Optimization Techniques:

- Reallocate bandwidth.
- Use CBR for VoIP, VBR for video, and ABR/UBR for data.
- Balance traffic across VCs.
- Upgrade overloaded switches.
- Implement traffic shaping.

Expected Outcome:

Reduced delay and jitter, improved voice/video quality, and efficient bandwidth utilization.

Question 3: MPLS Network Performance Issues

Root Causes:

- Congested Label Switched Paths (LSPs).
- Incorrect label mappings.
- Improper traffic engineering.
- Uneven bandwidth utilization.
- Routing inefficiencies.

Debugging Process:

- Verify MPLS label tables.
- Monitor LSP utilization.
- Trace packet paths.
- Check router configurations.
- Analyze latency and packet loss.

Optimization Techniques:

- Correct label mappings.
- Configure MPLS Traffic Engineering.
- Balance traffic across LSPs.
- Implement QoS.
- Optimize routing paths.

Expected Outcome:

Lower latency, better application performance, improved bandwidth utilization, and reliable communication.

Question 4: Error Detection and Correction Issues

Root Causes:

- Electrical interference.

- Signal attenuation.
- Network congestion.
- Weak error detection mechanisms.
- Undetected corrupted frames.

Debugging Process:

- Monitor error rates.
- Analyze frame corruption.
- Check network media.
- Identify congestion points.
- Test error control methods.

Optimization Techniques:

- Implement stronger CRC.
- Use Forward Error Correction (FEC).
- Apply Hamming Code.
- Improve cable shielding.
- Upgrade network equipment.
- Manage traffic efficiently.

Expected Outcome:

Higher data accuracy, fewer retransmissions, and improved reliability.

Question 5: Flow Control Problems

Root Causes:

- Sender transmits faster than receivers can process.
- Receiver buffer overflow.
- Excessive retransmissions.
- Network congestion.
- Inefficient flow control.

Debugging Process:

- Monitor buffer utilization.
- Analyze retransmissions.
- Measure throughput and latency.
- Check flow control settings.
- Identify congestion points.

Optimization Techniques:

- Implement Sliding Window Protocol.
- Use Dynamic Window Sizing.
- Apply TCP Flow Control.
- Increase buffer capacity.
- Implement congestion control.

- Prioritize examination traffic.

Expected Outcome:

Reduced packet loss, improved throughput, lower latency, and better bandwidth utilization.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated

Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation
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